

RAFFLES INSTITUTION
2013 Year 6 PRELIMINARY EXAMINATION

Higher 1



CANDIDATE NAME

CLASS

INDEX NUMBER

CHEMISTRY

8872/02

Paper 2

18 September 2013
2 hours

Candidates answer Section A on the Question Paper

Additional Materials: Answer paper
Graph paper
Data Booklet

READ THESE INSTRUCTIONS FIRST

DO NOT open this booklet until you are told to do so.

Write your name, class and index number in the spaces provided above.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (40 marks)

Answer **all** the questions.

Section B (40 marks)

Answer any **two** questions on separate answer paper.

You are to begin each question on a fresh sheet of paper.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

For Examiner's Use	
Paper 1	/ 30
Paper 2	
A1	/ 11
A2	/ 12
A3	/ 7
A4	/ 10
B5	/ 20
B6	/ 20
B7	/ 20
Total	/110

Section A (40 marks)

Answer **all** questions in the spaces provided.

For
Examiner's
Use

A1 This question is about the chemistry of magnesium, aluminium and calcium.

(a) (i) Write the full electronic configurations of magnesium and aluminium.

Mg

Al

(ii) The first ionisation energy of aluminium is less than that of magnesium. Explain.

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[2]

(b) Describe and explain how the acid-base behaviour of aluminium oxide, Al_2O_3 , differs from that of magnesium oxide, MgO . Write equations for the reactions you choose to illustrate your answer.

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[4]

(c) Magnesium trisilicate, $\text{Mg}_2\text{Si}_3\text{O}_x$, can neutralise hydrochloric acid, producing a magnesium salt, water and a crystalline solid with a giant covalent structure.

(i) Determine a value for x .

(ii) Using the value obtained in (c)(i), write an equation to show the reaction between magnesium trisilicate and hydrochloric acid.

.....

[2]

Iontophoresis is a technique which uses an electric current to deliver ions and drug through the skin. Small amounts of inorganic salts, such as molten calcium chloride and molten magnesium sulfate, can be used to aid drug delivery in iontophoresis.

(d) (i) Explain why molten calcium chloride and molten magnesium sulfate can be used to aid drug delivery through the skin when an electric current is applied.

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(ii) With reference to the *Data Booklet*, state and explain which ion is larger, Ca^{2+} or Mg^{2+} .

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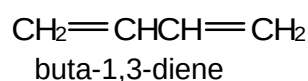
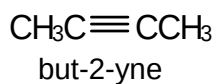
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[3]

[Total: 11]

- A2** But-2-yne and buta-1,3-diene are two unsaturated, gaseous hydrocarbons with the formula C_4H_6 .



The standard enthalpy change of formation for these two hydrocarbons are given below:

$$\Delta H_f^\ominus [\text{but-2-yne}] = +165 \text{ kJ mol}^{-1}$$

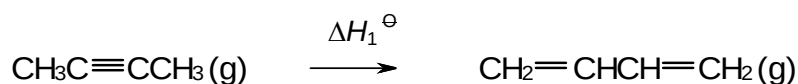
$$\Delta H_f^\ominus [\text{buta-1,3-diene}] = +110 \text{ kJ mol}^{-1}$$

- (a)** Define the term *standard enthalpy change of formation*.

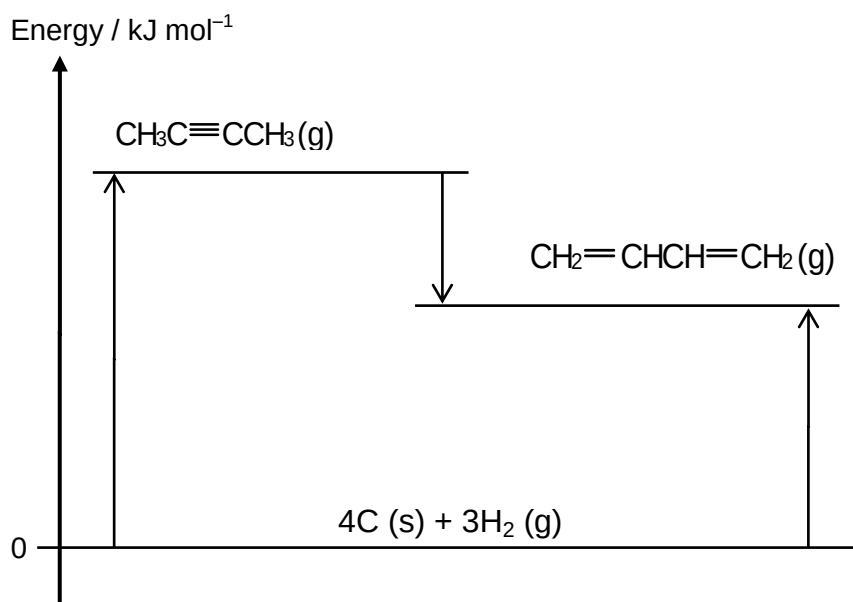
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[1]

- (b)** The following conversion between these two isomeric hydrocarbons is proposed.



- (i)** Label the values of standard enthalpy change of formation for the two hydrocarbons clearly, on the energy level diagram below.



- (ii)** Label $\Delta H_1^\ominus$ on the energy level diagram above and calculate its value.

- (iii) If the structure of a molecule allows for the delocalisation of bonding electrons, the molecule is likely to be more stable.

Based on this, state and explain the relative stability of buta-1,3-diene and but-2-yne.

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- (iv) Based on your answers above, state if the standard enthalpy change of combustion of buta-1,3-diene is more or less exothermic than that of but-2-yne.

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[6]

The standard enthalpy change of hydrogenation, $\Delta H^{\ominus}_{\text{hydrogenation}}$, is defined as the enthalpy change when one mole of an unsaturated compound reacts completely with excess hydrogen to form a saturated compound at 1 atm and 298 K.

- (c) (i) Write an equation, with state symbols, for the hydrogenation of but-2-yne.

- (ii) The condition needed for this hydrogenation is similar to that needed for the reaction between butene and hydrogen. State the condition needed for this reaction in (c)(i).

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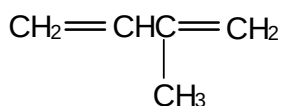
- (iii) Use bond energies from the *Data Booklet* to calculate the standard enthalpy change of hydrogenation of but-2-yne.

- (iv) Based on your answers above, determine the standard enthalpy change of formation of butane.

[5]

[Total: 12]

A3 Isoprene, the building block of Vitamin A, can undergo a wide range of reactions.



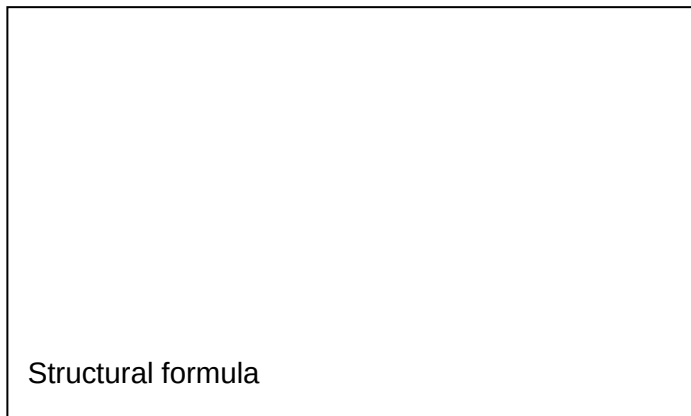
isoprene

- (a) Isoprene reacts with KMnO_4 under different conditions. Other than decolorisation of KMnO_4 , describe what you might observe in the following reactions. Give the structural formulae of the organic products formed.

- (i) Cold alkaline KMnO_4

Structural formula

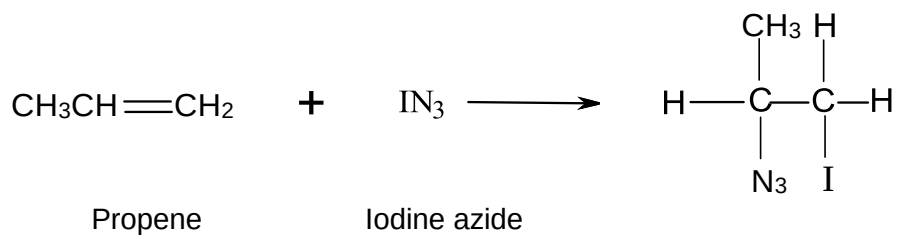
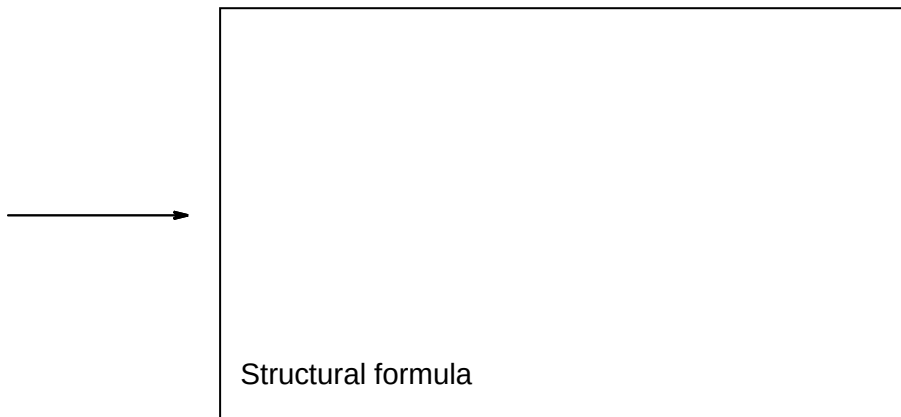
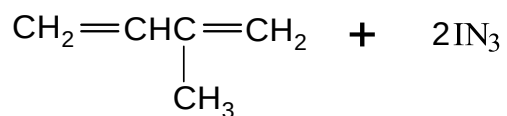
Observation:

(ii) Hot acidified KMnO_4 

Observation:

[4]

(b) Iodine azide is a covalent inorganic azide that can undergo electrophilic addition with alkenes in the following manner:

(i) Give the structural formula of the organic product formed from the reaction between isoprene and IN_3 .

- (ii) Another covalent inorganic azide is chlorine azide, ClN_3 . Explain which bond, Cl-N or I-N , has a higher bond energy.

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[3]

[Total: 7]

A4 Arsenic, As, is a toxic element present in the environment. It forms both inorganic and organic compounds.

- (a) Arsenic trioxide, As_2O_3 , has a melting point of 312°C . It reacts with alkalis.

Describe the structure and bonding of As_2O_3 .

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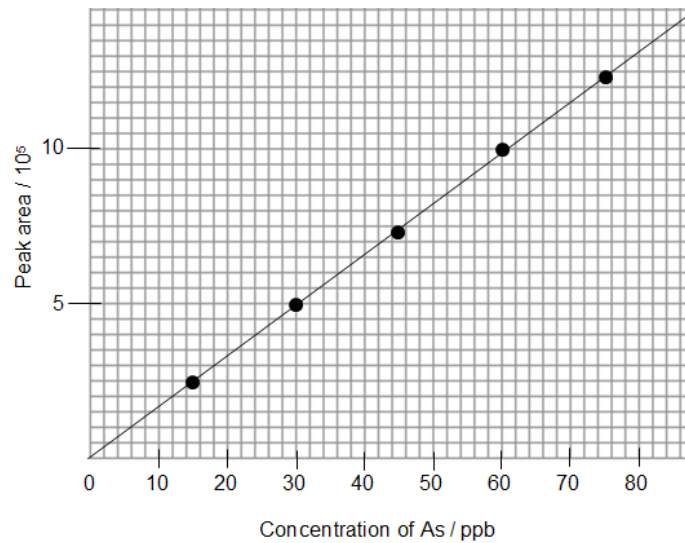
[1]

- (b) The oral LD_{50} , also known as median lethal dose, of As in humans is estimated to be 2 mg/kg . This indicates that 2 mg of As per kg of human-body-mass, if administered, is fatal.

Calculate the mass of As_2O_3 needed to be taken orally to put the life of a man, weighing 60 kg , into danger.

[2]

- (c) Measurements of the level of arsenic in urine is a biological indicator of exposure to arsenic. Urine samples are analysed by mass spectrometry to determine the concentration of As. The graph below shows the relationship between the area under the peaks in the mass spectra obtained from such samples, and the concentration of As.



A man suspected of excessive exposure to As was admitted into the hospital, and his urine sample was analysed 11 days after exposure.

- (i) The peak area obtained from analysing the man's urine sample is 2.0×10^5 . Using the graph above, determine the concentration (in ppb) of As in the urine sample 11 days after exposure.

Concentration of As in ppb =

- (ii) Using the equation below, calculate the original concentration (in ppb) of As in the urine sample on Day 0.

$$[\text{As}]_n = [\text{As}]_0 \times \left[\left(\frac{1}{2} \right)^{\frac{n}{4}} \right]$$

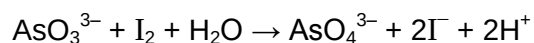
where, $[\text{As}]_n$ = concentration of As on n th day

$[\text{As}]_0$ = concentration of As on Day 0

n = number of days after exposure

[2]

- (d) Arsenite (AsO_3^{3-}) concentration can be found via iodometric titration with iodine as shown in the equation below.



- (i) Deduce the type of reaction, based on changes in oxidation number.

.....

- (ii) With the aid of a sketch of the Boltzmann distribution, explain how an increase in temperature would affect the rate of the above chemical reaction.

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[5]

[Total: 10]

Section B (40 marks)

Answer any **two** questions from this section on separate papers.

B5 This question concerns elements found in Period 3 of the Periodic Table.

(a) Pollution is the introduction of contaminants into a natural environment that causes harm to the ecosystem. Sulfur hexafluoride, SF_6 , is the most potent green house gas. In the presence of sparks and oxygen, SF_6 decomposes to give toxic sulfur oxyfluorides and sulfur tetrafluoride, SF_4 .

(i) Draw the structure of a molecule of sulfur tetrafluoride, SF_4 , showing the shape clearly.

(ii) State and explain the shape around the central atom in SF_4 .

(iii) Draw a diagram to show the intermolecular forces between SF_6 and water.

[5]

(b) Since the Industrial Revolution, emissions of sulfur dioxide and nitrogen oxides to the atmosphere have increased. The increase in sulfur dioxide and nitrogen oxides results in the production of acid rain which contains sulfuric(IV) acid and nitric(V) acid.

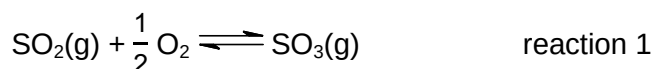
(i) Assuming there is an average of 156 tonnes of rainfall annually and HNO_3 is present in 1.5 parts per million (ppm) of the annual rainfall, calculate the concentration of HNO_3 , in mol dm^{-3} , present in the annual rainfall. [Given 1 tonne = 1.0×10^6 g, Density of HNO_3 = 1.0 g cm^{-3}]

(ii) A statue made of calcium carbonate is placed in the open. Write a balanced equation between calcium carbonate and HNO_3 .

(iii) Using your answer to **(b)(ii)** and assuming 0.1% of the annual rainfall came into contact with the statue, calculate the loss in mass of the statue in that year.

[4]

(c) In the manufacture of sulfuric acid, SO_2 is converted into SO_3 in the presence of a vanadium(V) oxide catalyst.



(i) Explain why a catalyst is needed for reaction 1.

(ii) State what will happen to the yield of SO_3 if the pressure of the system increases.

(iii) At 298 K, $\Delta H_f^\ominus (\text{SO}_2) = -297 \text{ kJ mol}^{-1}$ and $\Delta H_f^\ominus (\text{SO}_3) = -396 \text{ kJ mol}^{-1}$.

Use these data to calculate $\Delta H^\ominus$ for reaction 1.

[4]

- (d)** Phosphorus forms many useful chlorides and oxychlorides.

Phosphorus pentachloride, PCl_5 , is one of the most important chlorides, others being PCl_3 and POCl_3 . PCl_5 exists as white solid and is widely used as a chlorinating agent of organic compounds.

In the solid state, PCl_5 has an ionic lattice structure consisting of PCl_x^+ and PCl_6^- ions.

Deduce a likely identity and the shape of the cation, PCl_x^+ , and draw its dot-and-cross diagram.

[2]

- (e)** Sulfur is a source of some important oxychlorides. Thionyl chloride, SOCl_2 , and sulfuryl chloride, SO_2Cl_2 , are both colourless liquids at room temperature.

The phosphorus chlorides and sulfur oxychlorides react with water to produce acidic compounds.

- (i)** PCl_5 reacts with water in two separate ways.
 (1) 1 mole of PCl_5 reacts with 1 mole of H_2O to form a white solid, together with white fumes.
 (2) 1 mole of PCl_5 reacts with excess H_2O to form a very acidic solution.

Write equations, including state symbols, for these two reactions and explain the observations.

State the observation when a few drops of Universal Indicator is added to the resulting solution.

- (ii)** Both SOCl_2 and SO_2Cl_2 react vigorously with water.

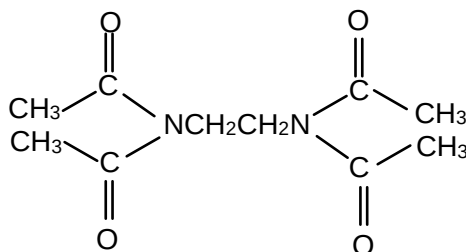
When 1 mole each of SOCl_2 and SO_2Cl_2 is added separately to excess water, 2 moles of HCl and a sulfur-containing product are formed in each reaction. The sulfur in the two compounds have different oxidation states.

Suggest the identities of the sulfur-containing products for each reaction and state the oxidation number of sulfur in each case.

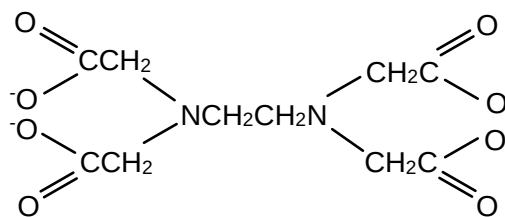
[5]

[Total: 20]

- B6 (a)** The salt of ethylenediaminetetraacetic acid, commonly known as EDTA^{4-} , and tetraacetythylenediamine, also known as TAED, are structurally similar to each other.



TAED

 EDTA^{4-}

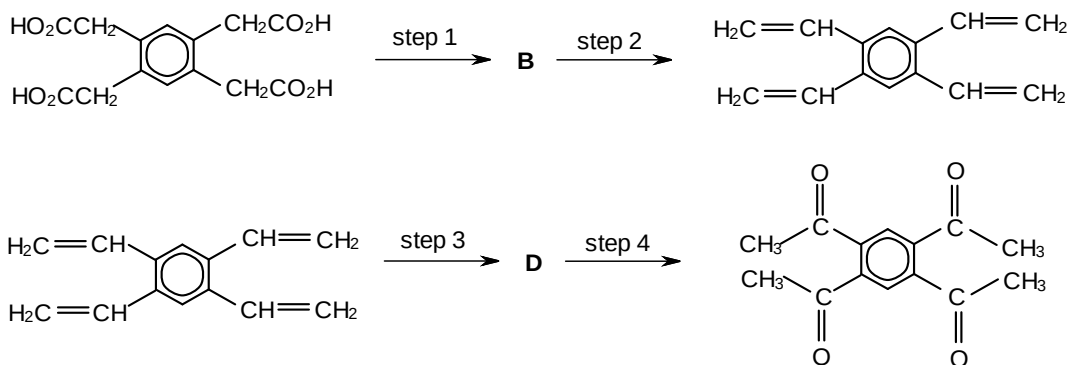
- (i)** Identify the functional group present in TAED.

EDTA^{4-} has been used to inhibit the catalytic decomposition of hydrogen peroxide. An example of such a catalyst is Mn^{2+} . EDTA^{4-} removes the Mn^{2+} catalyst by reacting with it to form a stable ion with co-ordination number six.

- (ii)** EDTA^{4-} forms dative bonds with Mn^{2+} . Explain why EDTA^{4-} is able to form bonds with Mn^{2+} .
- (iii)** Copy out the given formula of EDTA^{4-} and circle, on your diagram, the six atoms that form dative bonds with Mn^{2+} .
- (iv)** Hence, state the formula of the ion formed between EDTA^{4-} and Mn^{2+} .

[4]

- (b)** In examining reactions of analogous organic compounds, the following reaction scheme has been proposed.



State the reagents and conditions needed for the above conversions, identifying **B** and **D** formed.

[6]

(c) Compound **G**, $\text{C}_5\text{H}_9\text{O}_2\text{Cl}$, has the following characteristics.

- Effervescence is produced on adding sodium carbonate to **G**.
- It produces a yellow precipitate with hot aqueous alkaline iodine.
- It reacts with hot aqueous sodium hydroxide to give **J**.
- It does not decolourise hot acidified KMnO_4 .

Upon acidification, **J** forms compound **K**, $\text{C}_5\text{H}_8\text{O}_2$ in the presence of a small amount of hot concentrated sulfuric acid.

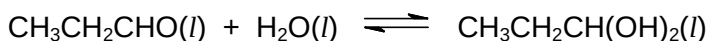
K has the following characteristics.

- It does not exhibit geometric isomerism.
- It does not decolourise aqueous bromine.

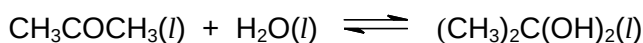
Deduce the structures of all the compounds, giving reasons for your answers.

[5]

(d) The reaction between propanal and water is a reversible reaction. The reaction is described as being in dynamic equilibrium.



- State the meaning of the term *dynamic equilibrium*.
- Write an expression for the equilibrium constant, K_c , for the reaction between propanal and water as shown above.
- At equilibrium, the concentrations of $\text{CH}_3\text{CH}_2\text{CHO}(l)$, $\text{H}_2\text{O}(l)$ and $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})_2(l)$ are $0.676 \text{ mol dm}^{-3}$, $0.676 \text{ mol dm}^{-3}$ and $0.324 \text{ mol dm}^{-3}$ respectively. Calculate the equilibrium constant, K_c , at 25°C , stating its units.
- Propanone, an isomer of propanal, also reacts with water in a reversible reaction as shown below.

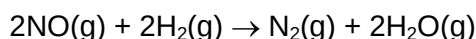


The numerical value for the equilibrium constant of this reaction was found to be 1.40×10^{-3} at 25°C . By using your answer to (c)(iii) and the above information, compare the relative reactivities of propanal and propanone.

[5]

[Total: 20]

- B7 (a)** Consider the reaction between hydrogen and nitrogen oxide at 800 K:



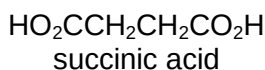
Experiments were carried out in an enclosed vessel to determine the effect of NO on the initial rate of experiment. This was done by varying the initial concentration of NO present in the system, while keeping the initial concentration of H_2 constant.

$[\text{NO}]^2 / \text{mol}^2 \text{ dm}^{-6}$	0.010	0.023	0.040	0.063	0.090
Initial rate / $\times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$	6.5	14.6	26.0	40.6	58.5

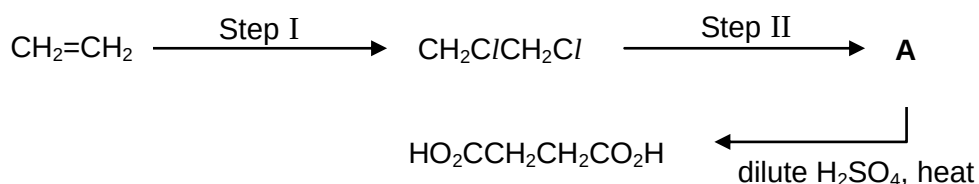
- Use the data provided above to plot the graph of initial rate against $[\text{NO}]^2$. Hence deduce the order of reaction with respect to NO.
- Given that the rate of reaction is directly proportional to the concentration of H_2 , write the rate equation for the reaction.
- Given that the initial concentration of H_2 for each experiment is 0.50 mol dm^{-3} , calculate the rate constant for the reaction, stating its units.
- Solid iron can be used to catalyse the reaction between NO and H_2 . Explain this by means of sketching, on the same axes, a clearly labeled reaction pathway diagram for the catalysed and uncatalysed reaction.

[7]

- (b)** Succinic acid, historically known as spirit of amber, plays an important biochemical role in the citric acid cycle. Its structure is shown below.

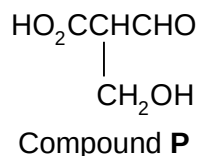


Succinic acid can be synthesized from ethene using the reaction scheme shown below.



- State the reagents and conditions used for steps I and II. Draw the structure of A.
- State the type of reaction taking place in step II, and write a balanced equation for the reaction.

- (iii) Compound **P** is a structural isomer of succinic acid:



Suggest a simple chemical test that could be used to distinguish compound **P** from succinic acid. State the reagents and conditions used and the expected observation for each compound.

[7]

- (c) (i) Calculate the pH of the $0.0100 \text{ mol dm}^{-3}$ hydrochloric acid solution.
- (ii) Calculate the pH of the $0.0100 \text{ mol dm}^{-3}$ 2-methylpropanoic acid solution, given that $pK_a = 4.88$.
- (iii) Explain the difference in the pH values calculated in (c)(i) and (c)(ii).
- (iv) 3-hydroxybutanoic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{COOH}$, has a pK_a value of 4.70. By considering their molecular structures, explain the difference in pK_a values between 3-hydroxybutanoic acid and 2-methylpropanoic acid.

[6]

[Total: 20]

END OF PAPER